

Severin Ihnat

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EDUCATION

Columbia University

Master of Science in Computer Science

New York, NY

May 2026

University of Rochester

Bachelor of Science in Biomedical Engineering

Rochester, NY

May 2024

Awards: Alpha Eta Mu Beta Honors Society (top 20% of BME class); Dean's List (all semesters)

TECHNICAL SKILLS

Languages: Python, Bash, SQL, C++ (basic)

ML / Robotics: PyTorch, OpenCV, computer vision, sensor/hardware integration

Tools / Infra: Git, Linux, Docker, REST APIs, WebSockets, InfluxDB, SLURM/HPC

Spoken: Ukrainian (advanced), French (basic)

EXPERIENCE

Columbia University

Computer Science Researcher

New York, NY

Sep 2024 – May 2026

- Built a real-time, automated ML pipeline on a SLURM GPU cluster, enabling a **500+ GPU-hour** parameter sweep with zero-touch job submission, result ingestion, and experiment tracking
- Extended a hierarchical sequence model with continual and active learning to study cross-domain generalization in protein-peptide binding across **6 domain pairs** and **1M+ training samples**
- Constructed and validated **10,000+ data points** from unstructured sources via an automated data-processing pipeline

NIDUS MB Technologies

Senior Design Lead

Rochester, NY

Dec 2023 – May 2024

- Led a team of **3** to automate manual cell extraction; built a real-time computer-vision pipeline (edge detection) that processed a live microscope camera feed to guide **closed-loop pipette positioning** — a full perception-to-actuation loop on physical hardware, with device control via a custom GUI

Regeneron Pharmaceuticals

Automation Core Intern (2023, 2024) & Protein Expression Intern (2022)

Tarrytown, NY

May 2022 – Aug 2024

- Developed HSL control scripts driving the Hamilton STAR liquid-handling robot to automate HEK293 cell-line development (splitting, cherry-picking, plate creation), integrating software with physical instrument hardware to cut hands-on time from **2 hrs to 30 min** per 96-well run
- Built a configurable Python GUI for user-defined serial-dilution patterns across 96/384-well plates; adopted by the Automation Core team across **10+ research experiments**

SELECTED PROJECTS

YOLOv8 Autonomous CV Training

Apr 2026

- Engineered an agentic hyperparameter-search loop that autonomously modifies, trains, scores, and git-commits or reverts YOLOv8 configurations, improving a weighted recall-precision score by **15.3%** across **20 controlled experiments** on the **VisDrone aerial-drone detection dataset** (UAV perception); found that augmentation degrades performance at low training scale

Smart Security Camera (ML)

Jun 2026

- Built a real-time computer-vision security camera in Python (YOLOv8 + ByteTrack) ingesting video from a webcam or a **Raspberry Pi 5 camera node** streaming MJPEG over HTTP, running multi-class detection with persistent cross-frame tracking and evaluating tracked objects against user-defined polygon zones via OpenCV
- Implemented loitering detection from per-object dwell time, auto-triggering clip capture (pre-trigger ring buffer + post-trigger frames to timestamped .mp4) and persisting structured zone entry/exit/loiter event logs for offline review

LEADERSHIP

- STEM Tutor**, UR Learning Center (CS, biology, BME); **Water Polo Captain**, UR Club Sports — led a 20+ person team